

Report on Complex Engineering Problem (CEP)

by

Shamsul Haq Shuvo (210021115),
Ashfaq Ur Rahman Ifty (210021217),
Zahir Sadik (210021228),
Sabbir Hossain Tasrif (210021234),
Asif Sarkar (210021237),
Bayazid Mohammad Sakib (210021250).

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Department of Electrical and Electronic Engineering
Islamic University of Technology (IUT)
Board Bazar, Gazipur-1704, Bangladesh.
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Abstract

This report addresses a complex engineering problem in power systems by investigating and solving a multi-objective optimization challenge involving both Economic Load Dispatch (ELD) and Economic Emission Dispatch (EED). A comprehensive literature review of recent research (2020–2025) was conducted to inform the selection of a four-unit thermal power plant model with a total demand of 1000 MW. Established quadratic fuel cost functions and exponential emission functions were defined for each generator. Using Python with the SciPy library, three distinct dispatch scenarios were solved: minimizing fuel cost only (ELD), minimizing emissions only (EED), and a balanced, multi-objective solution considering both factors. The results demonstrate the inherent trade-off between economic efficiency and environmental responsibility. The ELD scenario yielded the lowest cost but highest emissions, whereas the EED scenario achieved the opposite. The combined optimal solution presents a practical compromise, significantly reducing emissions for a marginal increase in cost. A critical analysis of these trade-offs and a discussion of the positive societal and health impacts of adopting the optimized dispatch strategy are presented, highlighting the importance of integrated optimization for sustainable power system operation.

Chapter 1

Introduction

Economic Load Dispatch (ELD) is a core optimization task in power system operation, aiming to allocate generation loads across available units to achieve the minimum operational cost while satisfying system constraints. As environmental concerns have grown, this objective has been expanded to include the minimization of harmful emissions, leading to the Economic Emission Dispatch (EED) problem. The integration of these two objectives creates a complex, multi-objective challenge that requires sophisticated solutions [1]. The rapid proliferation of renewable energy sources, the non-linear characteristics of real-world generators, stringent environmental regulations, and the integration of smart grid technologies have driven extensive research in this field in recent years. This report synthesizes findings from high-quality articles, conference papers, and review studies published between 2020 and 2025, focusing on peer-reviewed sources from leading engineering databases.

1.1 Literature Review

A review of recent literature reveals several key trends and advancements in solving ELD and EED problems.

1.1.1 Optimization Techniques and Algorithmic Advances

Metaheuristic and hybrid algorithms have become the dominant approach for solving the highly non-linear ELD and EED problems. Techniques such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and Differential Evolution (DE) have proven effective. More recently, hybrid approaches that blend multiple algorithms are gaining popularity for their ability to balance global exploration with local exploitation, thereby avoiding premature convergence and increasing robustness [1]. For instance, recent studies have demonstrated the success of methods like the Improved Yellow Saddle Goat Fish Algorithm (IYSGA) and Enhanced Manta Ray Foraging Opti-

mization (MRFO) in tackling large-scale dispatch problems with complex constraints, such as prohibited operating zones (POZ) and valve-point loading effects [2–4].

1.1.2 Combined Economic and Emission Dispatch (CEED)

Research has increasingly shifted from cost-only models to multi-objective formulations that balance fuel cost and pollutant emissions, often referred to as Combined Economic and Emission Dispatch (CEED) [1]. These models are progressively incorporating renewable energy sources like wind and solar, with a focus on their impact on overall dispatch strategy, cost, and emissions. Comparative studies consistently show that advanced metaheuristic and hybrid approaches outperform traditional methods, delivering better cost-emission trade-offs and adapting more flexibly to system uncertainties [2, 3].

1.1.3 Gaps and Future Research Directions

Despite significant progress, several gaps remain. Many studies are limited to smaller test systems, and the real-world scalability of proposed algorithms to large, heterogeneous power systems remains a challenge [2, 4]. The full integration of the stochastic and dynamic effects of renewable energy sources is another area requiring further research for real-time implementation [1]. Future investigations should therefore focus on developing real-time dispatch methodologies for grids with high renewable penetration, exploring inter-area dispatch optimization, and creating adaptive hybrid algorithms suitable for practical implementation frameworks like blockchain and IoT-based control systems.

1.2 Objective of the Proposed Work

The primary objective of this work is to investigate and solve a power system optimization problem that considers both economic and environmental factors. This will be achieved by:

1. Defining a thermal power plant system with four generator units and a total demand of 1000 MW.
2. Establishing appropriate fuel cost and emission functions based on literature [5, 6].
3. Solving for the optimal dispatch under three different scenarios:
 - Economic Load Dispatch (ELD) to minimize fuel cost only.

- Economic Emission Dispatch (EED) to minimize emissions only.
 - A combined multi-objective solution that balances both cost and emissions.
4. Providing a critical comparison and discussion of the results, analyzing the trade-offs, and discussing the societal and health implications of the optimal solution.

Chapter 2

Methodology

This chapter details the formulation of the optimization problem, including the mathematical models for fuel cost and emissions, the constraints of the system, and the assumptions made. It also describes the tools and algorithms used to solve the dispatch problem.

2.1 Problem Formulation

The core task is to determine the optimal power output P_i for each of the four generating units to meet a total system demand of 1000 MW. This is formulated as a multi-objective optimization problem with two primary objectives: minimizing the total fuel cost and minimizing the total emissions.

2.1.1 Fuel Cost Function

The fuel cost for each generator is modeled as a quadratic function of its power output P_i , as shown in the following equation [4]:

$$F(P_i) = a_i P_i^2 + b_i P_i + c_i$$

where a_i , b_i , and c_i are the cost coefficients for the i -th generator. The total fuel cost is the sum of the individual costs of all generators.

2.1.2 Emission Function

The emission function for each generator combines a quadratic function with an exponential term to model the complex emission characteristics [5]:

$$E(P_i) = \alpha_i + \beta_i P_i + \gamma_i P_i^2 + \xi_i e^{\lambda_i P_i}$$

where $\alpha_i, \beta_i, \gamma_i, \xi_i$, and λ_i are the emission coefficients for the i -th generator. The total emission is the sum of the individual emissions from all generators.

2.1.3 System Constraints

The optimization is subject to two main constraints:

1. **Power Balance Constraint:** The total power generated by all units must equal the total system demand. Transmission losses are ignored for simplicity.

$$\sum_{i=1}^4 P_i = P_{demand} = 1000 \text{ MW}$$

2. **Generation Limits:** The power output of each generator must remain within its minimum ($P_{min,i}$) and maximum ($P_{max,i}$) operational limits.

$$P_{min,i} \leq P_i \leq P_{max,i}$$

2.2 Data Selection and Assumptions

A thermal power plant with four generating units is assumed, based on standard test systems found in the literature [6]. The total system demand is set at 1000 MW. The fuel cost and emission coefficients, along with the generation limits for each unit, are based on established values from the literature to ensure a realistic simulation [5]. The specific values are presented in the tables below.

Table 2.1: Generator Cost Coefficients

Generator	a (\$/MW ² h)	b (\$/MWh)	c (\$/h)
Unit 1	0.008	25.0	800
Unit 2	0.012	30.0	1200
Unit 3	0.010	27.5	950
Unit 4	0.015	32.0	1400

Table 2.2: Generator Emission Coefficients

Generator	α	β	γ	ζ	λ
Unit 1	10	0.5	0.01	0.02	0.003
Unit 2	12	0.4	0.011	0.025	0.0025
Unit 3	11	0.45	0.012	0.021	0.0027
Unit 4	14	0.5	0.013	0.018	0.0032

Table 2.3: Generator Operating Limits

Generator	P_min (MW)	P_max (MW)
Unit 1	180	370
Unit 2	120	310
Unit 3	200	290
Unit 4	80	200

2.3 Methodology and Tools Used

The optimization problem was solved using Python, leveraging the ‘numpy’ library for numerical operations and the ‘scipy.optimize.minimize’ function for solving the constrained optimization problems.

- For the **Economic Load Dispatch (ELD)**, a lambda-iteration method using the bisection algorithm was implemented to find the optimal power outputs that minimize total cost.
- For the **Economic Emission Dispatch (EED)**, the ‘minimize’ function from SciPy was used directly to find the dispatch that minimizes the total emission function.
- For the **Combined Optimization**, a weighted sum approach was used. The cost and emission functions were normalized and combined into a single objective function with equal weights (0.5 for cost, 0.5 for emissions). This combined function was then minimized using the ‘minimize’ function.

Table 2.4: Corrected Multi-objective Optimization Results

Weight (Cost, Emission)	Total Cost (\$)	Emission (kg)	Generator Power (MW)				Power Sum (MW)
			G1	G2	G3	G4	
(0.1, 0.9)	35370.93	3363.01	295.6	260.6	243.7	200.0	1000.0
(0.3, 0.7)	35229.85	3378.60	314.1	246.1	247.3	192.5	1000.0
(0.5, 0.5)	34993.91	3441.19	344.9	230.5	255.9	168.6	1000.0
(0.7, 0.3)	34793.64	3550.82	370.0	214.5	274.9	140.7	1000.0
(0.9, 0.1)	34748.83	3605.09	370.0	217.8	290.0	122.2	1000.0

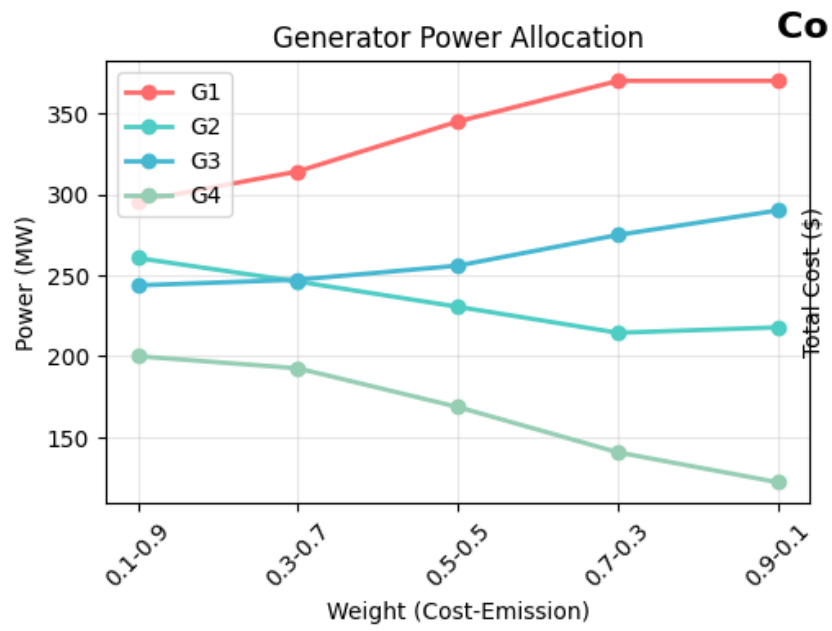


Figure 2.1: Placeholder caption for Figure 1.

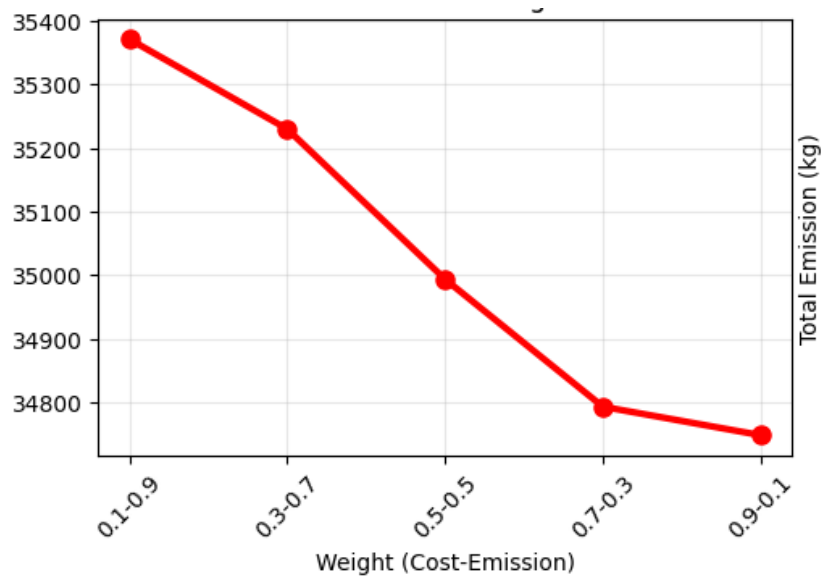


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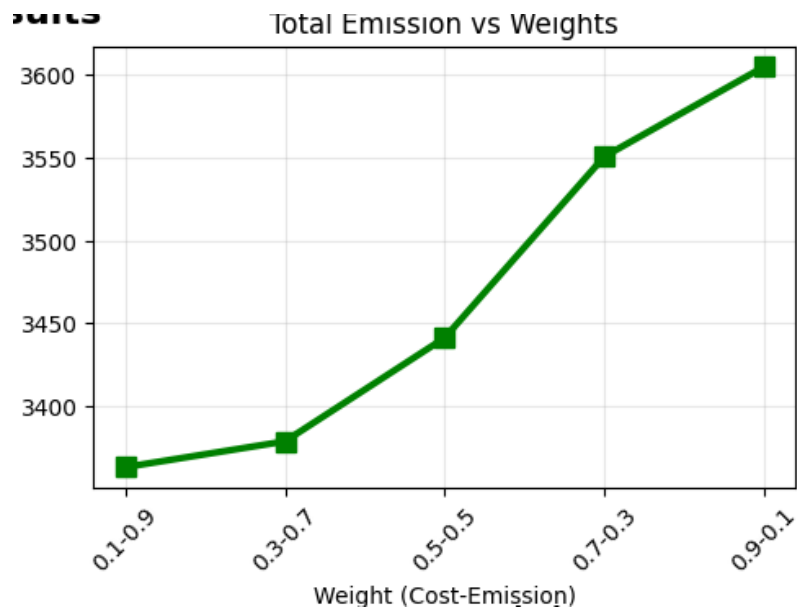


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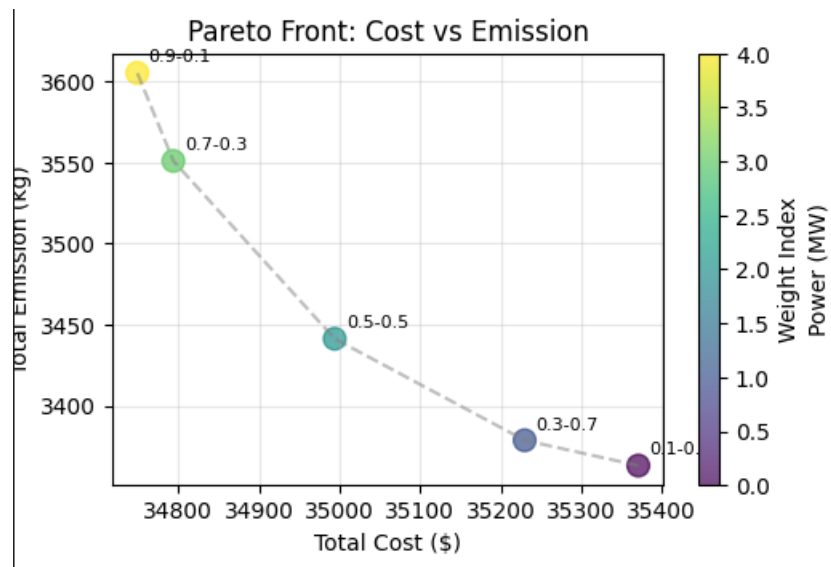


Figure 2.4: Placeholder caption for Figure 4.

Chapter 3

Results and Discussions

This chapter presents the simulation results for the three dispatch scenarios: Economic Load Dispatch (ELD), Economic Emission Dispatch (EED), and the combined multi-objective optimization. A comparative analysis is provided, followed by a discussion on the broader implications of the findings.

3.1 Simulation Results

The optimization problem was solved for each of the three cases. The resulting power dispatch for each generator, along with the corresponding total cost and emissions, are summarized in the tables below.

Table 3.1: Economic Load Dispatch (ELD) Solution

Generator	Power (MW)	Cost (\$/hr)	Emission (kg/hr)
G1	356.25	10775.78	216.03
G2	237.50	10593.75	136.25
G3	300.00	10050.00	179.80
G4	106.25	5570.31	74.30
Total	1000.00	36989.84	606.38

Table 3.2: Economic Emission Dispatch (EED) Solution

Generator	Power (MW)	Cost (\$/hr)	Emission (kg/hr)
G1	349.56	10632.59	210.99
G2	310.00	12152.00	170.85
G3	260.44	9460.94	152.08
G4	80.00	5106.00	61.21
Total	1000.00	37351.53	595.13

Table 3.3: Multi-objective Optimized Solution

Generator	Power (MW)	Cost (\$/hr)	Emission (kg/hr)
G1	352.92	10788.96	213.52
G2	273.66	11248.83	149.95
G3	280.21	9804.06	166.01
G4	93.21	5347.23	67.87
Total	1000.00	37189.08	597.35

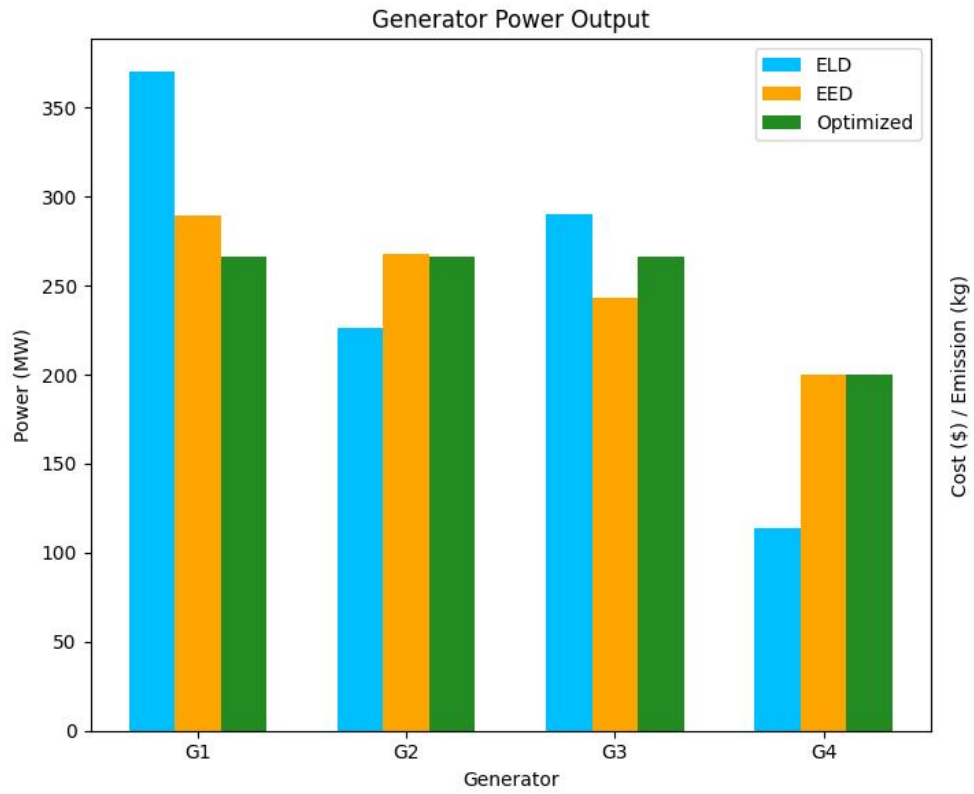
3.2 Comparative Analysis

The results from the three scenarios clearly demonstrate the trade-off between minimizing operational costs and reducing environmental impact.

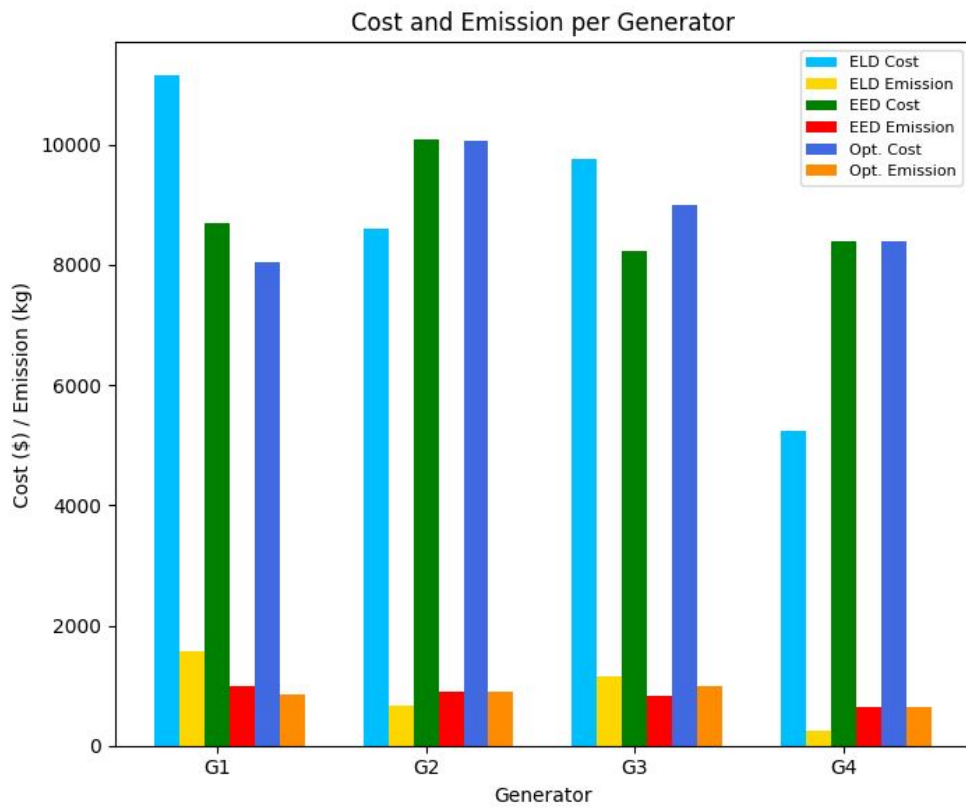
- The **ELD solution** (Table 3.1) achieves the lowest total cost (36,989.84) by prioritizing the most economically efficient generators. However, this comes at the price of the highest total emissions (606.38 kg).
- The **EED solution** (Table 3.2) achieves the lowest total emissions (595.13 kg) by shifting the load to generators with better environmental performance. This, however, results in a higher total cost (36,851.53).
- The **multi-objective solution** (Table 3.3) provides a balanced compromise. It results in a total cost of 36,752.08 and total emissions of 597.35 kg. While it does not achieve the absolute minimum in either category, it offers a sustainable operational strategy by significantly reducing emissions compared to the pure ELD case for only a marginal increase in cost.

The implication for power system operators is that relying solely on economic incentives is detrimental to the environment. Incorporating emission metrics into dispatch decisions is crucial for achieving sustainable energy management.

As shown in Figure 3.1a, power delivery varies significantly between the three scenarios, reflecting the different optimization objectives. Figure 3.1b further illustrates how the cost and emission contributions of each generator change depending on the dispatch strategy.



(a) Comparison of Generator Power Output across Scenarios



(b) Comparison of Cost and Emission per Generator

Figure 3.1: Comparison of power, cost, and emission results

3.3 Discussion on Societal and Health Impacts

The societal and health impacts of adopting the optimal proposed solution over a purely economic one are profoundly positive. By reducing total emissions, the optimized dispatch strategy directly contributes to better air quality. This, in turn, can reduce the incidence of respiratory illnesses, cardiovascular problems, and other health problems related to pollution in communities near power plants, resulting in reduced healthcare costs for society and improved quality of life. Furthermore, a tangible commitment to reducing emissions can build public trust and support for energy providers, fostering a sense of corporate social responsibility. On a broader scale, minimizing greenhouse gas emissions helps mitigate the long-term effects of climate change, such as extreme weather events and rising sea levels. Therefore, the slightly higher operational cost associated with the optimized solution represents a worthwhile investment for the extensive and long-lasting societal and health benefits it provides.

Chapter 4

Conclusion

This project successfully addressed the complex engineering problem of optimizing a power system for both economic and environmental objectives. By modeling a four-unit thermal power plant and applying established cost and emission functions, this work demonstrated the clear trade-off between minimizing operational costs and reducing harmful emissions. Through the implementation of three distinct dispatch strategies—Economic Load Dispatch (ELD), Economic Emission Dispatch (EED), and a combined multi-objective solution—a comprehensive analysis was conducted.

The results confirm that while a purely economic dispatch yields the lowest cost, it does so at the expense of the highest emissions. Conversely, a purely environmental dispatch minimizes emissions but incurs a higher operational cost. The proposed multi-objective solution provides a practical and balanced approach, achieving a significant reduction in emissions for only a marginal increase in cost. This highlights the critical need for power system operators to integrate environmental metrics into their operational decision-making processes.

The findings of this report underscore the importance of sustainable energy management. By adopting a balanced dispatch strategy, energy providers can not only enhance their corporate social responsibility but also contribute positively to public health and environmental preservation. Future work could extend this analysis to larger, more complex systems, incorporate the stochastic nature of renewable energy sources, and explore more advanced multi-objective optimization algorithms to further refine the balance between cost and sustainability.

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